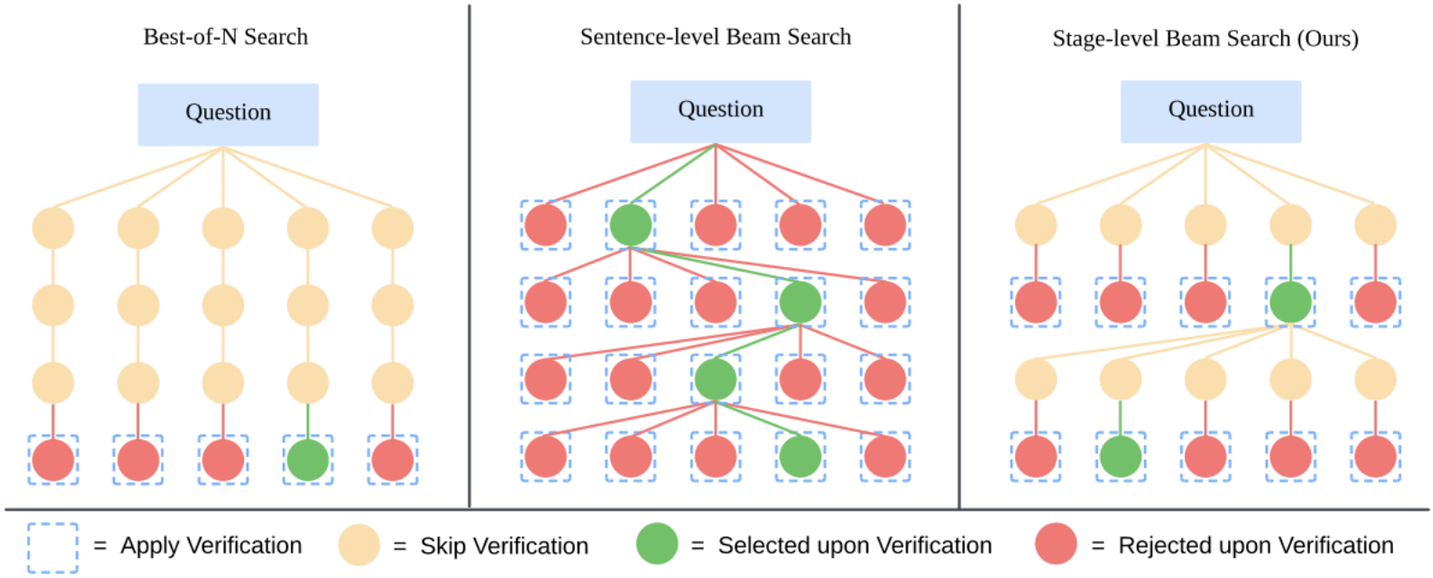

Figure 4. **An illustration of inference approaches.** Best-of-N search generates N complete responses and selects the best one among them; Sentence-level Beam Search generates multiple candidate options for each sentence and chooses the best one. In contrast, our Stage-level Beam Search generates candidates for each reasoning stage (e.g., summary, caption, reasoning, and conclusion) and selects the best option at each stage. Best-of-N search operates at a coarse level, while Sentence-level Beam Search is overly granular, and our method achieves an optimal balance and achieves the best performance.

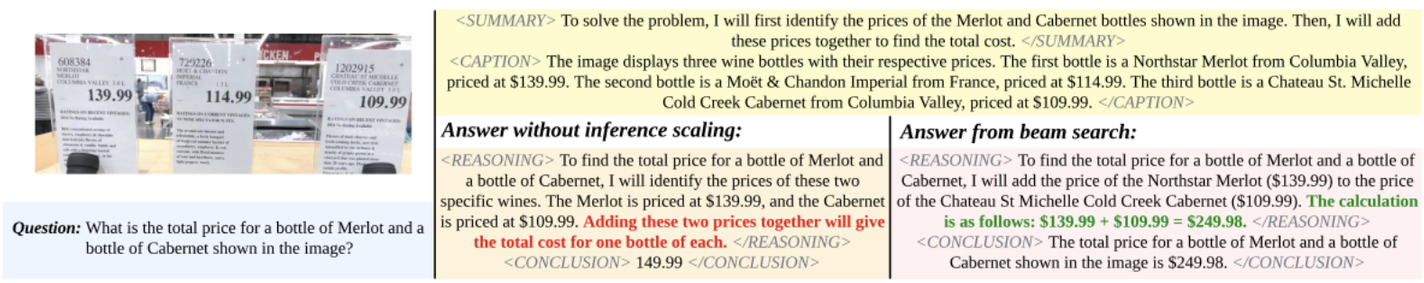

Figure 5. **Comparison of `LLaVA-o1` performance with and without stage-level beam search.** Our stage-level beam search is effective in selecting better reasoning during model inference.

[23] and **ScienceQA** [34], which target scientific questions. **CLEVR-Math** [13], an extension of CLEVR, focuses on arithmetic analysis in visual contexts. Table 1 shows the number of QA pairs selected from each dataset.

Model Training. The `LLaVA-o1-100k` dataset we construct can be used to further conduct Supervised Fine-Tuning (SFT) on any existing model to enhance reasoning capabilities. In this work, we select the Llama-3.2-11B-Vision-Instruct [40] model as the base model, and perform a full parameter fine-tuning by using the `LLaVA-o1-100k` dataset. The training is conducted on a single node with 8 H100 GPUs.

3.2. Effective Inference Time Scaling using Stage-level Beam Search

After training, our objective is to further enhance the model's reasoning ability during inference. Specifically, we leverage the stage-based outputs of `LLaVA-o1`, which provides an ideal granularity for inference time scaling. Our method follows the steps below:

- Sample N responses for the first stage in the solution.
- Randomly sample 2 responses and let the model determine which is better, keeping the better response.
- Repeat for $N-1$ times, retaining the best response.
- Sample N responses for the next stage, then repeat steps 2-4 until all stages are processed.

Notably, **it is the structured output design of `LLaVA-o1` that makes this approach feasible**, enabling efficient and accurate verification at each stage. This validates the effectiveness of structured output in improving inference time scaling. An illustration of the three approaches is shown in Figure 4.

We provide an example in Figure 5. When inference time scaling is not applied, although the model generates correct reasoning steps, it fails to arrive at a concrete answer during the reasoning process. This causes the model to make a guess in the conclusion phase, leading to an incorrect result. In contrast, with inference time scaling, the model retains the reasoning steps leading to the final result, ensuring the correctness of the answer.